

Replicated Calibration of an ESP32 Soil-Moisture Sensor Using Controlled Water Addition

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Study type: descriptive, replicated sensor-calibration experiment

Abstract

This study examined how controlled additions of 0–40 g of water affected the analog-to-digital converter (ADC) output of a soil-moisture sensor connected to an ESP32. Three experimental rounds produced 11 independently prepared cup samples and 55 formal displayed ADC values. For the main replicated levels, 0 g, 20 g, and 40 g each had three independently prepared samples. Mean ADC decreased monotonically from 3308.8 at 0 g to 1871.7 at 20 g and 1112.2 at 40 g. Exploratory 10 g and 30 g samples extended the curve, with means of 2433.6 and 1397.4. A threshold of 2200 separated every recorded 0 g reading from every recorded reading at 20 g or more, although one of five 10 g readings crossed below the threshold. These results support the threshold as a practical local setting for this sensor, substrate, and protocol, while also showing that the 10 g region is a transition zone requiring further replication.

1. Research Question

How does added-water mass (0–40 g) affect ESP32 soil-moisture sensor ADC readings across independently prepared samples, and how reliably does a threshold of 2200 separate dry samples (0 g) from more-watered samples (20–40 g) under this experimental setup?

2. Hypothesis

As added-water mass increases, the ADC reading will decrease. Dry 0 g samples are expected to remain above 2200, while samples receiving 20 g or more are expected to remain below 2200.

3. Materials

- One ESP32 development board and one soil-moisture sensor
- Laptop with Thonny and MicroPython, USB cable, and jumper wires
- Identical plastic cups and the same type of dry growing medium
- Water and a device capable of measuring water mass in grams
- Ruler, labels, marker, and materials for cleaning/drying the sensor

4. Method

1. Three experimental rounds were completed. Round 1 tested 0, 20, and 40 g; Round 2 tested 0, 10, 20, 30, and 40 g; Round 3 independently repeated 0, 20, and 40 g.
2. For each round, new dry growing medium was placed in identical cups to approximately the same amount and height, and the cups were labelled by water condition.
3. The specified water mass was added, the medium was mixed, and samples were left for approximately 10 minutes to distribute the water.
4. The same sensor was used sequentially for all samples. It was inserted vertically to a marked depth of approximately 4 cm and cleaned and dried between samples.
5. The ESP32 read the sensor through GPIO33 with 11 dB attenuation. The program averaged five rapid ADC conversions taken at 100 ms intervals to create each displayed ADC value.
6. Five formal displayed values were recorded per cup while maintaining the same insertion depth and repositioning the probe as consistently as practicable.
7. A loose wire initially produced constant 4095 readings. These diagnostic values were excluded before formal data collection because variable readings returned after reconnection.

Measurement unit clarification. Each formal value is an average of five rapid conversions. The dataset therefore contains 55 formal displayed values, but only 11 independently prepared cup samples; the rapid conversions and five within-cup readings are technical repeats, not independently prepared samples.

4.1 Experimental Design

Round	Water levels	Independent cups	Formal readings
1	0, 20, 40 g	3	5 per cup (15)
2	0, 10, 20, 30, 40 g	5	5 per cup (25)
3	0, 20, 40 g	3	5 per cup (15)
Total	Five water levels	11	55 displayed values

Table 1. Experimental design across the three rounds.

4.2 Statistical Analysis

For each cup, the mean, minimum, maximum, range, and sample standard deviation (SD) were calculated from the five formal displayed values. For 0, 20, and 40 g, condition means and between-sample SDs were then calculated from the three independent cup means. The 10 and 30 g levels each had one cup, so between-sample SD could not be estimated. A linear fit across the five condition means was used descriptively; it was not treated as a universal physical calibration equation.

5. Results

Water (g)	Independent samples (n)	Mean ADC	Between-sample SD	Sample-mean min-max	Threshold result
0	3	3308.8	114.2	3216.2–3436.4	15/15 above
10	1	2433.6	Not estimated	2433.6–2433.6	4/5 above; 1/5 below
20	3	1871.7	33.7	1839.0–1906.4	15/15 below
30	1	1397.4	Not estimated	1397.4–1397.4	5/5 below
40	3	1112.2	40.9	1067.0–1146.6	15/15 below

Table 2. Condition-level summary. SD values use independent cup means; SD is not estimated where $n = 1$.

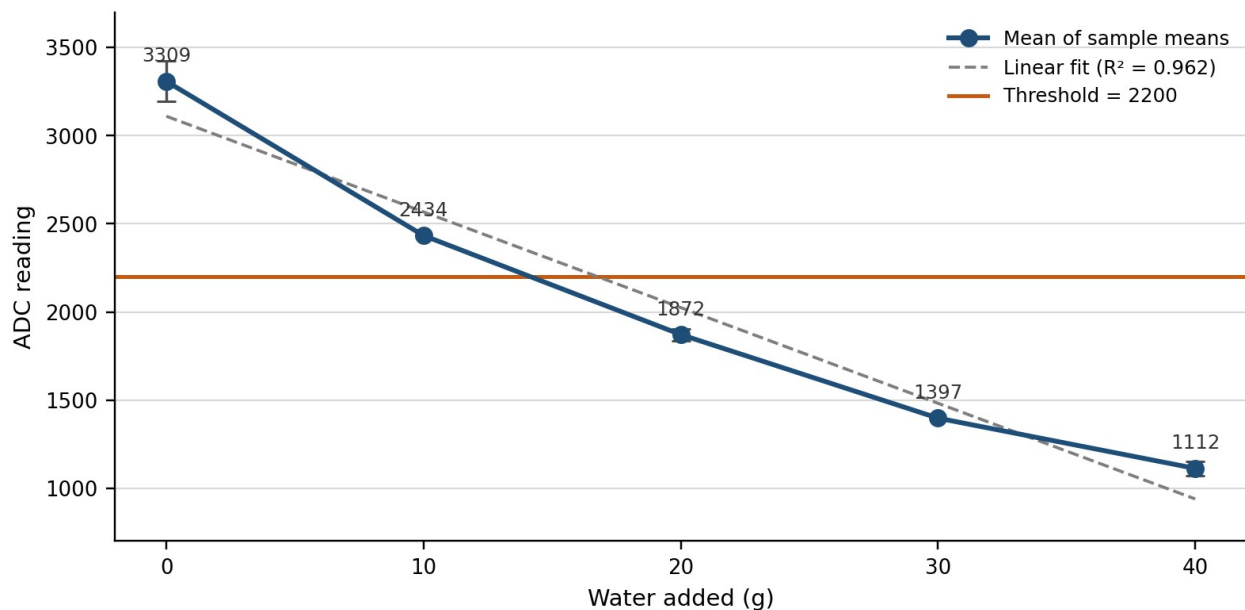


Figure 1. Mean ADC by added-water mass. Error bars show between-sample SD for the replicated 0, 20, and 40 g levels ($n = 3$ cups each); 10 and 30 g each had one cup. The dashed grey line is a descriptive linear fit.

5.1 Main Findings

- ADC decreased monotonically across the five tested condition means: 3308.8 (0 g), 2433.6 (10 g), 1871.7 (20 g), 1397.4 (30 g), and 1112.2 (40 g).
- Across the independently prepared 0, 20, and 40 g cups, between-sample SDs were 114.2, 33.7, and 40.9 ADC units, respectively. This shows that the large separation between water levels was reproduced across rounds.
- Every 0 g displayed value was above 2200, and every displayed value at 20–40 g was below 2200. The 10 g cup was close to the decision boundary: four readings were above 2200 and one reading (2052) was below it.
- Across the five condition means, the descriptive linear fit was $\text{ADC} = 3110.6 - 54.3 \times \text{water (g)}$, with $R^2 = 0.962$.
- Before measurement, the 30 g mean had been predicted as approximately 1496 ADC from the initial 20 and 40 g results. The observed 30 g sample mean was 1397.4, an absolute error of 98.6 ADC (7.1% of the observed mean).

5.2 Sample-Level Descriptive Statistics

Sample	Round	Water (g)	Mean	Min	Max	Within-sample SD
R1-0g	1	0	3273.8	3057	3620	235.5
R1-20g	1	20	1869.6	1806	1923	51.2
R1-40g	1	40	1123.0	1037	1208	69.6
R2-0g	2	0	3216.2	2997	3402	171.6
R2-10g	2	10	2433.6	2052	2600	218.1
R2-20g	2	20	1839.0	1706	1936	104.3
R2-30g	2	30	1397.4	1243	1503	113.0
R2-40g	2	40	1067.0	956	1209	105.7
R3-0g	3	0	3436.4	3025	3784	303.2
R3-20g	3	20	1906.4	1799	2007	94.2
R3-40g	3	40	1146.6	1035	1307	109.1

Table 3. Descriptive statistics for each independently prepared cup sample (five formal displayed values per sample).

6. Discussion

The expanded experiment supports the original hypothesis and strengthens the initial pilot study. The same ordering—0 g highest, 20 g intermediate, and 40 g lowest—appeared in all three independently prepared rounds. The intermediate 10 and 30 g conditions also followed the expected downward pattern, giving a more informative calibration curve than the original three-point test.

The 2200 threshold performed consistently for the operational distinction tested here: dry 0 g samples versus samples receiving at least 20 g of water. However, the 10 g sample straddled the threshold, so the system could switch decisions when the soil is near this transition region. A practical implementation should therefore use additional replication around 10–20 g and may benefit from hysteresis or a requirement for several consecutive readings before switching the pump.

The condition means show a strong descriptive linear relationship over the tested range, but the relationship should not be assumed to remain linear for every substrate, sensor, or water range. ESP32 ADC response can also vary with attenuation, device characteristics, and reference voltage; official documentation recommends careful ADC configuration and calibration when accurate voltage conversion is required [1,2]. This study therefore uses raw ADC counts as a reproducible local control signal rather than claiming a universal moisture percentage.

7. Limitations

- Only one sensor and one growing-medium type were tested, so the calibration is specific to this hardware and substrate.
- The 0, 20, and 40 g levels had three independently prepared cups, but the 10 and 30 g levels had only one cup each; their uncertainty cannot yet be estimated across samples.
- Five formal readings within each cup were technical repeats. They improve measurement stability but do not replace additional independently prepared cups.
- Growing-medium mass was controlled operationally by using the same amount and height, but exact dry mass and gravimetric or volumetric water content were not independently measured.
- Water distribution, sensor contact, insertion angle, supply voltage, temperature, long-term drift, and repeated wetting/drying cycles were not systematically tested.
- A loose signal connection initially caused saturation readings of 4095. These diagnostic values were excluded, and the incident demonstrates the importance of checking wiring before collection.

8. Conclusion

Across three rounds, increasing added-water mass produced lower ESP32 soil-moisture ADC readings. The main 0, 20, and 40 g conditions were independently replicated three times and maintained clear separation. Under this setup, a threshold of 2200 correctly separated all recorded dry 0 g readings from all recorded readings at 20 g or more. The 10 g sample identified a transition region near the threshold, so further replication at 10–20 g is the most useful next experiment. The study is therefore a valid small-scale replicated calibration research project, while its threshold remains a local engineering setting rather than a universal soil-moisture standard.

9. References

- [1] MicroPython. "Quick reference for the ESP32: ADC." [MicroPython ESP32 Quick Reference](#)
- [2] Espressif Systems. "ADC Calibration Driver." [Espressif ESP-IDF ADC Calibration documentation](#)

Appendix A. Experimental Evidence



Figure 2. Round 1 samples prepared with 0 g, 20 g, and 40 g of added water.



Figure 3. Sensor marked to maintain an insertion depth of approximately 4 cm.

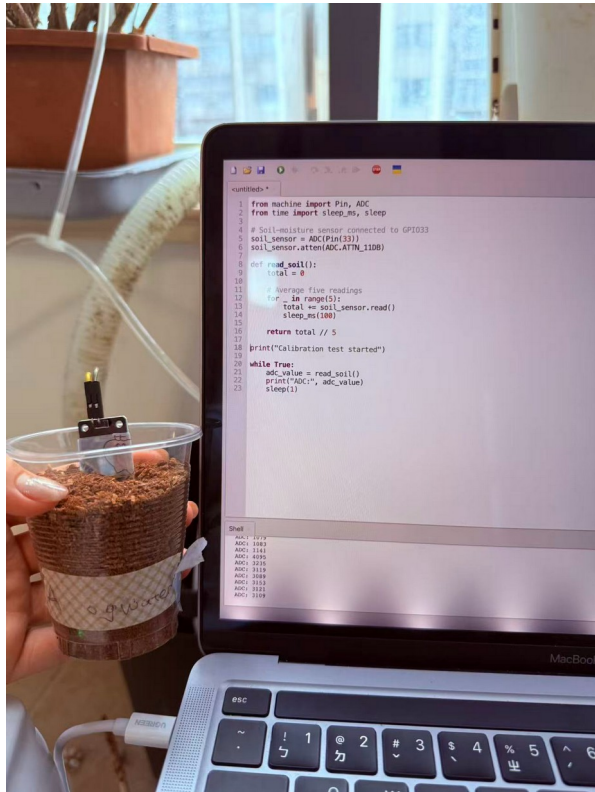


Figure 4. Round 1 measurement for the 0 g sample.

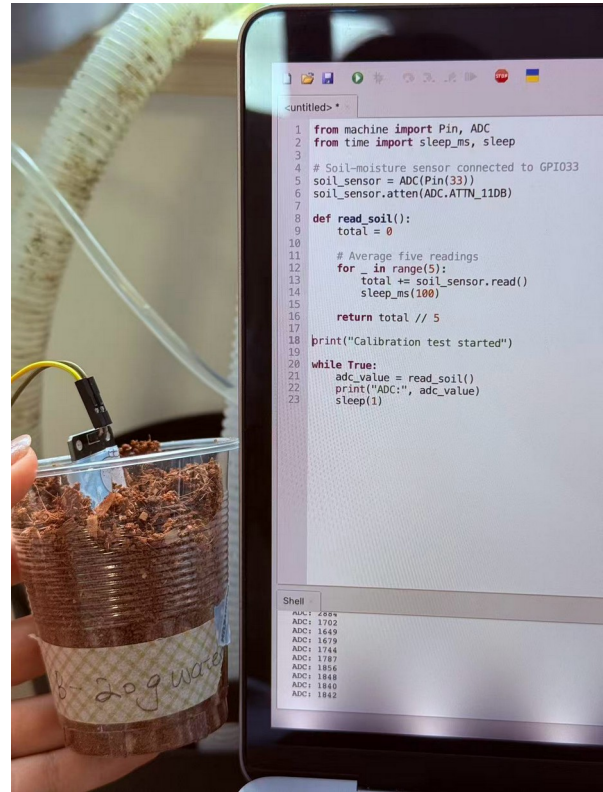


Figure 5. Round 1 measurement for the 20 g sample.

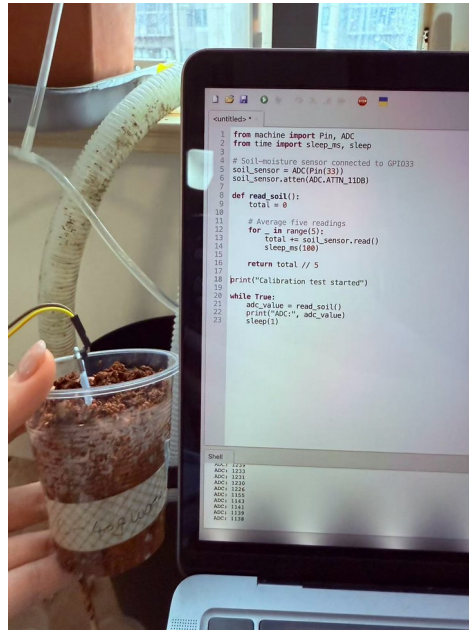


Figure 6. Round 1 measurement for the 40 g sample.



Figure 7. Round 3 independently prepared 40 g, 20 g, and 0 g samples (left to right), photographed under indoor lighting.

Figure 7 documents the actual indoor setup used for the third experimental round; the condition labels, relative fill levels, and sensor placement are visible.

Appendix B. Calibration Code

The following MicroPython code read the sensor through GPIO33. Each displayed ADC value was the integer average of five rapid conversions collected at 100 ms intervals.

```
from machine import Pin, ADC
from time import sleep_ms, sleep

soil_sensor = ADC(Pin(33))
soil_sensor.atten(ADC.ATTN_11DB)

def read_soil():
    total = 0

    for _ in range(5):
        total += soil_sensor.read()
        sleep_ms(100)

    return total // 5

print("Calibration test started")

while True:
    adc_value = read_soil()
    print("ADC:", adc_value)
    sleep(1)
```

Appendix C. Data Availability

The complete source code, 55-row raw dataset, statistical workbook, and project documentation are available in the GitHub repository: [Sensor-Driven Smart Watering System](#)

Recommended repository files: data/soil_moisture_calibration_raw_data.csv and data/soil_moisture_calibration_results.xlsx. The report statistics can be reproduced from those data.